RAKSHAM · AMAZON AI CYBER SECURITY HACKATHON · IIT DELHI

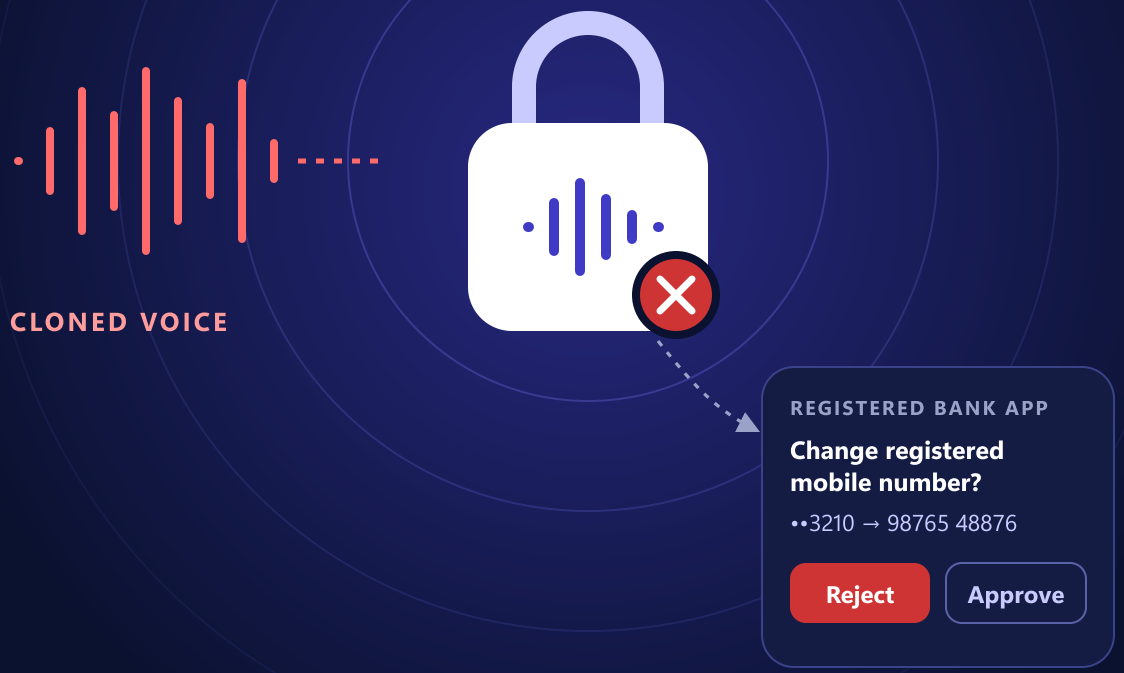
PS-01 · Voice Cloning Fraud Detection | Bank contact centres

# EchoLock

## A copied voice can't unlock a bank account.

EchoLock runs inside a bank's contact centre. It flags cloned speech on the bank's own call line, and holds every sensitive account change until the real customer approves **that exact change** in their registered bank app. A stolen voice alone can't take over the account.

| PRIMARY USER | HARMFUL MOMENT | OWNER |
|---|---|---|
| **Contact-centre agent & fraud analyst** | **Caller asks to change account data** | **Bank contact-centre ops + fraud risk** |




TEAM
**RapidActionForce**

**Delhi Technological University (DTU)**
Tushar Singh Bisht · Reedhan Garg · Shobhit Pandey · Shikhar Singh

CODE, DATA & RESULTS
**github.com/tusharsinghbisht/raksham-iitd-ps1**

02 THE PROBLEM

# Your voice has become a copyable password, and banks still trust it on the phone.

**3 sec** of audio was enough for an **85% voice match** with free cloning tools, in McAfee's tests.

**47%** of Indian adults surveyed had faced an AI voice scam or knew someone who had. That's **nearly 2× the global 25%**.

**83%** of Indian victims in that survey **lost money**.

Source: McAfee, *The Artificial Imposter* (2023), survey of 7,054 adults in 7 countries.

## HOW A VOICE GETS STOLEN

- ▶ Instagram reels & YouTube
- 🎙 Forwarded voice notes
- ☎ "Wrong number" calls
- 🎤 Interviews & meetings

→ **Open-source cloning**
TTS or real-time voice conversion. Free, runs on a laptop in minutes.

→ **Sounds like you**
on an 8 kHz phone line, where the flaws are hardest to hear

## WHAT A PHONE-BANKING CALL RELIES ON TODAY, AND WHY EACH ONE FAILS

### Caller ID
Can be spoofed with VoIP. A familiar number is not the customer.

### Security Qs
DOB, address, PAN and recent payments leak through breaches and social media.

### The voice
A clone is made to sound like the account holder, and can beat voice-ID checks too.

### Agent instinct
Agents handle hundreds of calls under time targets. Nobody can hear a good clone.

⚠ The phone channel is where banks **trust a voice the most** and **check it the least**. That's the gap EchoLock closes.

03 THREAT SCENARIO · USER · HARM

# The attack we stop: a fraudster clones the customer's voice and calls the bank to change their account details.

▼ ECHOLOCK INTERVENES HERE

1 **Harvest the voice**
30 s of the customer's speech from a reel or a call

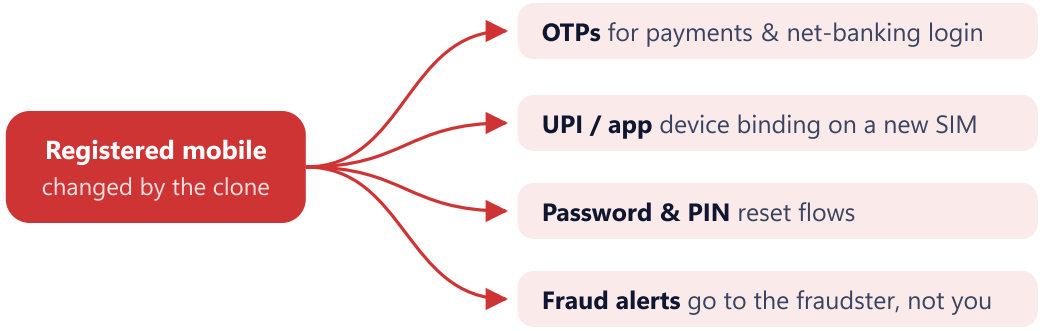
2 **Clone it**
TTS or live voice conversion, plus leaked KYC answers

3 **Call the bank as them**
Spoofed caller ID; passes security questions easily

4 **Change the data**
"I lost my phone, please update my mobile number"

5 **Take over the account**
OTPs now reach the fraudster. Reset, re-bind, drain.

## WHY CHANGING DATA IS THE REAL PRIZE




Change one field and the attacker gets every control built on it. The real customer often learns days later.

## SENSITIVE REQUESTS WE HOLD

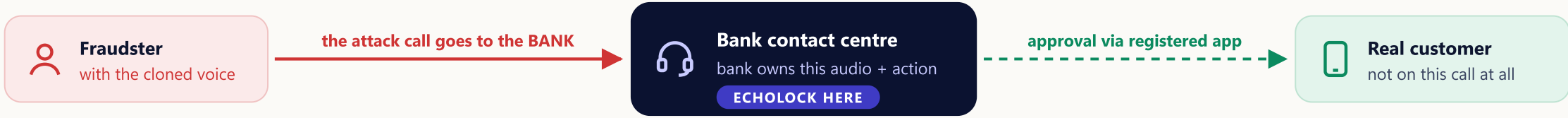
- Registered mobile / email change
- New device or SIM binding
- Transfer / UPI limit increase
- Address change + card reissue
- Add beneficiary / nominee
- Unblock card or account

## WHO'S INVOLVED

- **Primary user: contact-centre agent.** Has to decide while the caller is on the line.
- **Fraud analyst.** Reviews flagged cases and needs evidence, not a bare score.
- **Protected person: the account holder.** Not on the call and has no idea it's happening.

04 WHERE TO DEFEND

# Why we guard the bank's contact centre, not the customer's phone.




The customer's phone **never hears the clone**, so a phone app has nothing to analyse. The bank does.

## Detecting on the customer's phone

Not possible, and wouldn't help with this attack

- **Third-party apps can't capture call audio.** Android limits `VOICE_CALL` capture to system apps (`CAPTURE_AUDIO_OUTPUT`). Google Play banned Accessibility-API call recording in May 2022, and iOS has no call-audio API.
- **Wrong place for this attack.** The fraudster calls the bank, not the victim.
- **Privacy red flag.** Recording or uploading every personal call clashes with the DPDP Act and Play policy.
- **Coverage gaps.** Protection would depend on millions of installs, phone models and OS versions. Feature-phone users get nothing.

## Detecting at the bank contact centre

Legitimate audio, the right moment, one integration

- **The bank owns the call line.** Audio already passes through its IVR and agent systems under existing call-recording notices. No phone permissions needed.
- **The bank owns the action.** The change request lands here, so we can hold it *before* it runs.
- **The bank owns the duty.** RBI's 2025 authentication directions allow risk-based additional checks. That's our hook.
- **One deployment covers every customer**, on any phone, with an independent registered channel back to the real person.

**What about fake "bank official" calls to customers?** That's a different attack: the fraudster calls the customer, and phone apps still can't hear the call. It's better solved with verified-caller signals (TRAI 1600-series numbers, an in-app "Is my bank calling me?"). We list it as a future extension, not our PS-01 scope.

05 CURRENT JOURNEY → ECHOLOCK JOURNEY

# Today a voice that sounds right gets the change. With EchoLock, only the real customer can approve it.

**TODAY** A cloned caller asks to change the registered mobile number

**Cloned voice calls**
spoofed caller ID,
calm and convincing

→

**Agent asks security Qs**
DOB, address,
last transaction

→

**Clone answers them all**
leaked data plus a
voice that sounds right

→

**Agent updates mobile**
no signal that anything
is wrong

→

**Account taken over**
OTPs redirected,
savings drained

**WITH ECHOLOCK** Same call, but nothing changes without independent approval

**Same cloned call**
EchoLock listens on the
bank's line, no extra step

→

**Agent opens request**
"Change mobile" goes
on hold automatically

→

**Voice risk ELEVATED**
agent sees: "possible
impersonation, keep pending"

→

**Exact change → app**
real customer sees the full
new no. "98765 48876?"

→

**"I never asked" → Reject**
request cancelled,
case sent to analyst

### Customer

Stays safe even if a clone fools the agent. A genuine request costs one tap in an app they already use.

### Agent

Gets a plain next step instead of a raw "87% fake" score. No need to accuse the caller.

### Fraud analyst

Gets a ranked case with reason codes, score timeline and approval status, ready for a callback.

### Bank

Fewer account takeovers, an audit trail per change, and checks in line with RBI's risk-based guidance.

06 SOLUTION & DIFFERENTIATION

# Three layers: Detect, Confirm, Review. If one fails, the others still stop the attack.

LAYER 1

## Detect cloned speech

- A **pretrained self-supervised anti-spoofing model** (wav2vec 2.0, NII AntiDeepfake) runs on the bank's call audio.
- Scores **4-second speech windows**, tuned for narrowband phone audio.
- Returns an honest **UNKNOWN** when audio is too short or too noisy. It never guesses "genuine".

LAYER 2

## Confirm the exact change

- Every sensitive change is sent to the customer's **already-registered bank app**.
- The app shows the **exact action**, including the **full new number** ("••3210 → 98765 48876"). Signed, single-use, 60 s expiry.
- A clone can copy a voice. It can't copy a **registered device plus biometric unlock**.

LAYER 3

## Human review with evidence

- A flagged voice opens a **case even after an app approval**. Contact and limit changes also wait a **24-hour cooling-off**, so a coached customer has time to cancel.
- The analyst sees reason codes, a score timeline and approval status, then **calls back on the registered number**.
- The **bank keeps final authority**. The model never runs an action itself.

### BUILT TO SURVIVE DETECTOR FAILURE

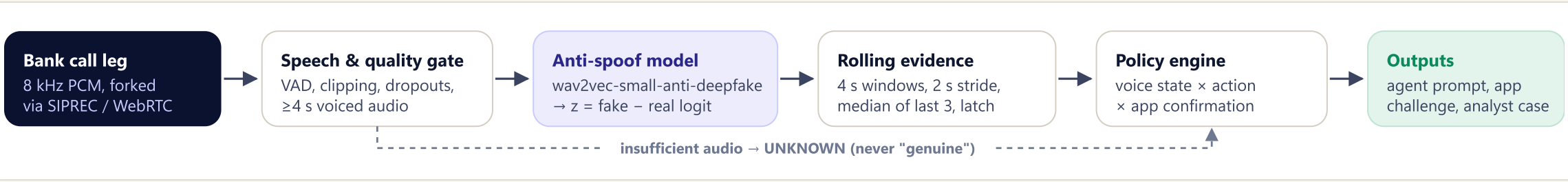
| SITUATION | VOICE LAYER | APP LAYER | OUTCOME |
|---|---|---|---|
| **Clone, detected** | Elevated | Real customer rejects | **Blocked + analyst case** |
| **Clone, missed** | Not elevated | No approval arrives | **Change never happens** |
| **Genuine customer** | Not elevated | Approves in ~10 s | **Proceeds to bank checks** |
| **Genuine, uses TTS aid** | Elevated | Approves | **Quick review, no penalty** |
| **Real customer, coached** | Not elevated | Approves under pressure | **24 h hold + alerts; one tap cancels** |

### HOW WE DIFFER

| TYPICAL DETECTOR | ECHOLOCK |
|---|---|
| "Is this voice fake?" | **"Did the account holder approve this exact change?"** |
| Score → auto-block | Score → more checks, never an accusation |
| Tested on clean lab clips | Tested on phone-codec, Indic speech |
| One model failure = loss | Missed clone still can't change data |
| Penalises TTS users | Same non-voice path for everyone |

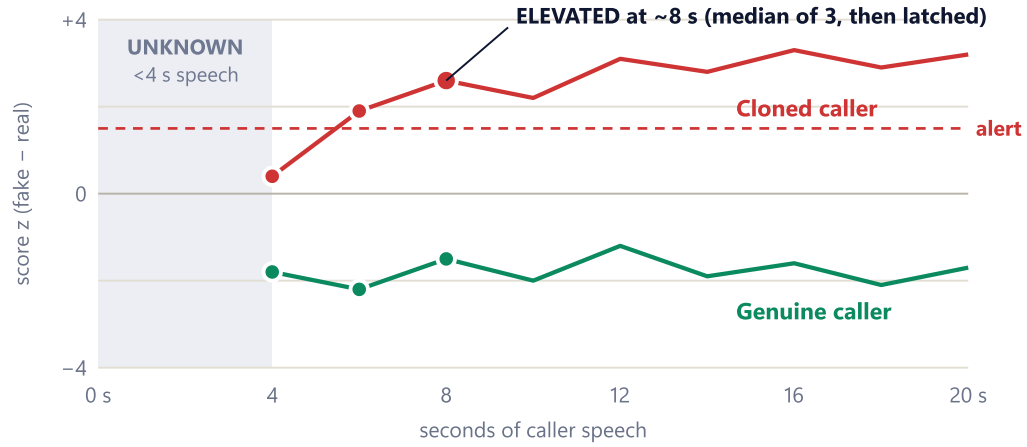
07 CORE DETECTION & DECISION LOGIC

# Rolling evidence from one strong model, gated by quality, with a decision policy that can't fail open.




## HOW THE VOICE STATE EVOLVES DURING A CALL

Illustrative until pilot results




## DECISION POLICY FOR SENSITIVE CHANGES

| VOICE STATE | APP APPROVES | REJECTS / TIMES OUT |
|---|---|---|
| Not elevated | Eligible for normal bank checks | Cancel; offer callback |
| Unknown | Eligible; flag "no voice evidence" | Cancel; assisted recovery |
| Elevated | **Hold for analyst review** (possible coercion) | **Cancel + priority case** |

### RULES THAT OVERRIDE THE MODEL

- Action and **call context** (number match, SIM swap, repeats) come from bank systems.
- The confirmation prompt starts **at once**. It never waits on the detector.
- No freeze, blacklist or report from **a voice score alone**.
- **"Lost my phone" / no app:** call can only **protect** (block card, freeze). Re-register via branch, video-KYC or net banking + 24 h hold.
- $\tau$ is set **per codec** on a speaker-disjoint dev split at 1% false alarms, then **frozen**.

08 VALIDATION PLAN & FIRST RESULTS

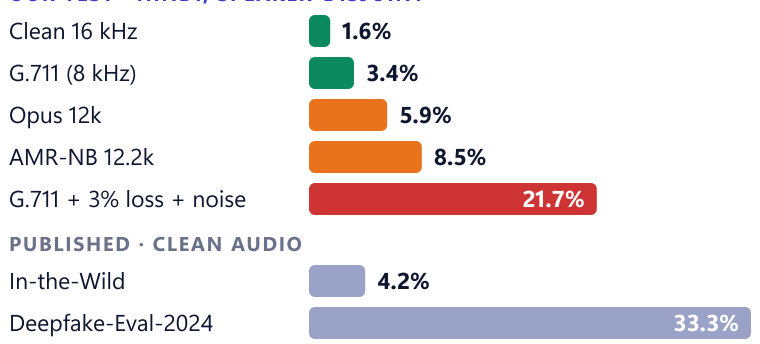
# We tested it on phone-quality Hindi calls, not clean lab clips.

## OUR "BHARAT-PHONE" TEST SET

1. **Public Hindi data, real vs cloned**
   Real Kathbath speech vs IndicSynth clones of the same speakers. Neither is in the model's training data.
2. **2 clone types + our own voices**
   XTTS-v2 voice-cloning TTS and FreeVC voice conversion, plus consented team recordings in Hindi, English and Hinglish.
3. **Phone-channel stress**
   8 kHz resample, G.711 / AMR-NB / Opus codecs, packet loss, noise, speakerphone replay.
4. **Leak-free splits**
   Speaker- and session-disjoint before augmentation. Thresholds frozen before the test set is opened.

## WHY THE CONFIRM LAYER IS ESSENTIAL

Error rate (EER) of **wav2vec-small**, the exact model we run:




Phone codecs keep errors under 9%, but **packet loss + noise pushes it to 22%**. So τ is set **per codec**, poor audio is UNKNOWN, and app confirmation carries the safety guarantee.

Ours: 960 test clips, 70 speakers. Kathbath real vs IndicSynth clones (XTTS-v2, FreeVC), call-level. Published: NII model card.

## WHAT WE FOUND · DETAILS IN A·1–A·3

- **1,727 clips × 6 channels** scored; 22 ms per window on a laptop GPU
- **τ must be per codec:** it moves from −2.27 (G.711) to +1.13 (clean)
- **XTTS-v2 clones are harder** than FreeVC on every channel
- On our own voices: **1.7% false alarms clean, 16% on G.711**, so τ is re-set on each bank's calls in shadow mode
- **26/26 confirm-layer tests pass**: replay, tamper, expiry, wrong key
- *Still to measure:* UNKNOWN rate, time-to-alert, analyst load

## KEY ASSUMPTIONS

**A1** · Banks can fork call audio to an internal service (SIPREC is standard). *Validate with a contact-centre stakeholder.*

**A2** · Most customers have a registered app. Others go through a staffed callback on the registered number.

**A3** · *Verified:* the model runs locally, 22 ms per 4 s window on a laptop GPU, 0.3 s on CPU.

**A4** · *Measured:* loss + noise raises errors to 22%. The design stays safe because Layer 2 doesn't rely on audio.

**Done so far**
and next

✓ **Reproducible model run**: our PyTorch loader, checkpoint rev 9a13264, timed on named hardware

✓ **Results per channel**, with sample counts and 95% intervals, never a lone accuracy number

**Next · timed end-to-end demo**: live XTTS clone call to app rejection to analyst case

09 PRIVACY, SECURITY & RESPONSIBLE AI

# If we're wrong, the customer gets a verification step, never an accusation.

## Minimal data

**WE PROCESS**
**Call audio:** a ≤16 s rolling buffer in memory, cleared when the call ends. Nothing new is archived.

**WE STORE (FLAGGED CASES ONLY)**
Score timeline, reason codes, model version, approval status. Encrypted, 7-day TTL.

**WE NEVER COLLECT**
Voiceprints, transcripts, contacts, OTP / PIN / card data, or anything from the customer's phone.

## When the model is wrong

**False alarm** A genuine caller just approves in the app. The analyst releases with a reason. No freeze, no "fraud" label.

**Missed clone** Still needs the real customer's app approval, so the change can't run.

**Unknown** Poor audio or a worker outage falls back to the normal secure route. It never fails open to "voice OK".

**Fairness** TTS and speech-impaired users take the same tap path, with no disability disclosure. False alarms audited per language, gender and channel.

## Misuse & adversaries

- **Adaptive attackers:** codec and noise augmentation; Layer 2 holds even if the model is beaten.
- **Replay / tampering:** nonce, action digest, 60 s expiry, single use.
- **Coached customers:** app asks "Is anyone on a call telling you to do this?" A yes goes to review. Contact and limit changes wait 24 h, with alerts and one-tap cancel.
- **Score forging:** only authenticated model workers can submit scores.
- **No surveillance:** no bulk API and no public cloning service. Consent-only test clones.

**0** **voiceprints** enrolled or stored, ever

**≤16 s** **of audio** held at once, in memory only

**7 days** **max retention** for flagged-case evidence (pilot default)

**0** **automatic freezes** or reports from a voice score

A human decides before any high-impact action, and **the bank stays the authoriser**. EchoLock only recommends. Not solved by design: malware on a stolen unlocked phone, or a compromised bank core.

10 DEPLOYMENT PATHWAY & 48-HOUR PLAN

# From a 48-hour demo to a bank shadow pilot, without touching customers until it's proven.

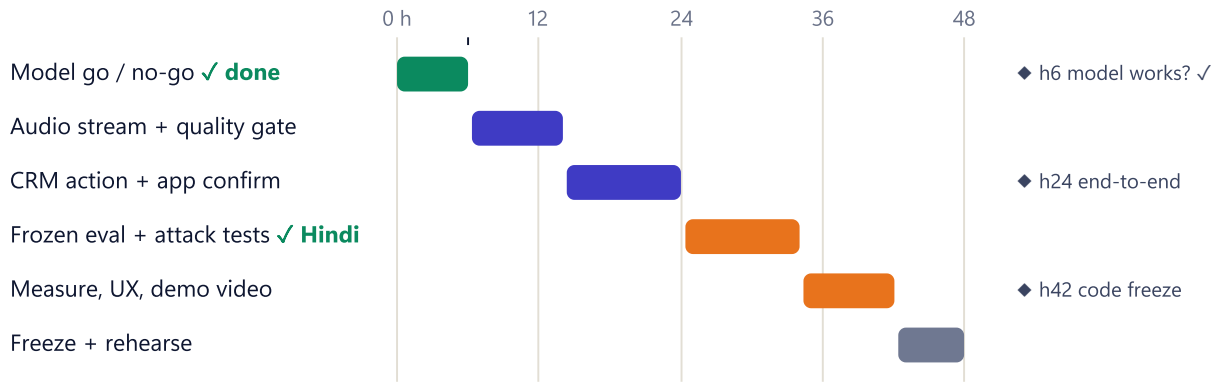
1. **48-h prototype**
   consented audio, mock CRM + app
2. **Shadow mode, one queue**
   scores logged, no customer impact
3. **Agent assist + app confirm**
   staffed recovery, measured friction
4. **All sensitive queues**
   licensed model, drift monitoring

**Owner:** contact-centre operations with fraud-risk and infosec teams.

**Integration:** SIPREC/WebRTC audio fork · REST policy API · CRM agent widget · push SDK in the existing bank app.

**Operational load:** about 1 MB RAM per live call. Overload returns UNKNOWN, never voice-only approval. Model rollback keeps bank auth intact.

## 48-HOUR FINALE BUILD




## LIVE DEMO WE'LL SHOW JUDGES

1. A consented clone calls and asks to **change the registered mobile**
2. The agent screen goes **ELEVATED** in about 8 s
3. The phone app shows the exact change. **Reject** → cancelled, case opened
4. The genuine speaker approves in one tap. A TTS user gets no penalty
5. A replayed or tampered approval is **refused**

### CUT LINE IF HOURS SLIP

- **Keep:** detector on recorded clips
- **Keep:** action hold + app confirm
- **Keep:** analyst view + replay test
- **Drop 1st:** live WebRTC
- **2nd:** UI polish
- **3rd:** extra codec sweeps

**Hackathon cost: about ₹0.** Open-source model and tools on our own laptops. Roles: ML & data · backend & security · frontend & media · validation & product.

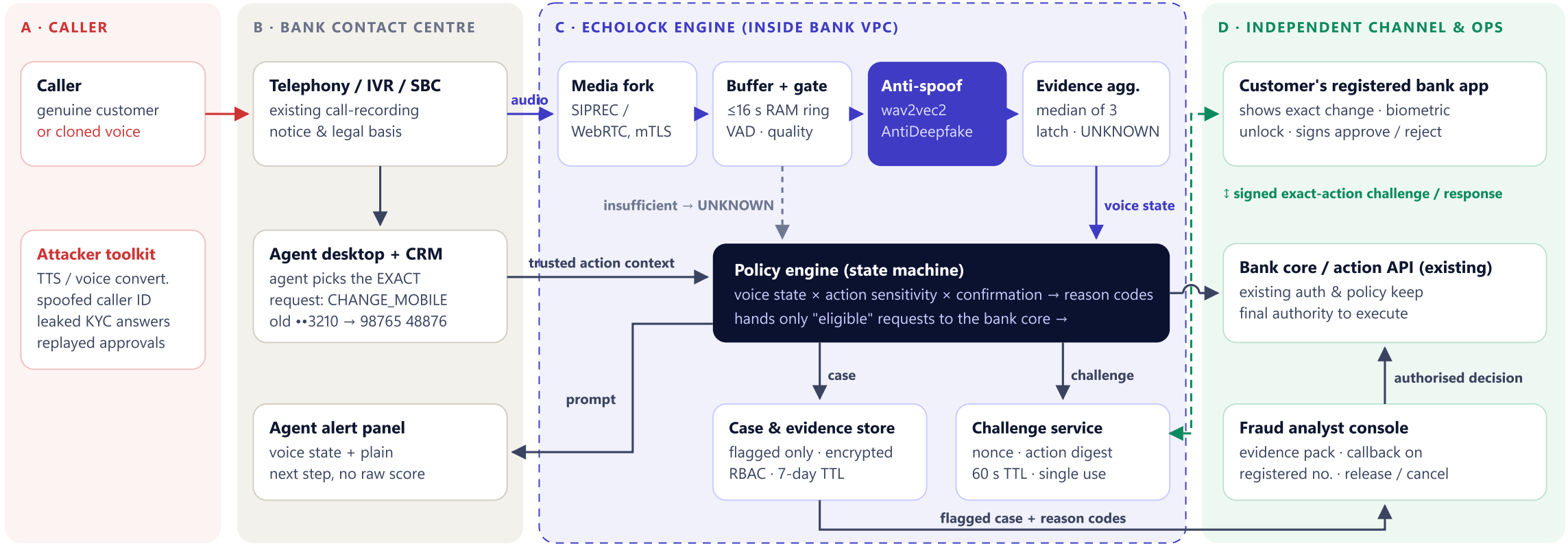
**Pilot cost estimate:** only calls with a sensitive request are scored, so cost scales with those calls, not all traffic. We expect one 8-vCPU server (about ₹20–25k/month on-demand cloud) to cover a pilot queue. *Assumption, to be measured in shadow mode.*

# EchoLock — Technical Architecture

PS-01 Voice Cloning Fraud Detection · RAKSHAM, IIT Delhi × Amazon · github.com/tusharsinghbisht/raksham-iitd-ps1

**Team RapidActionForce** · DTU · Tushar Singh Bisht · Reedhan Garg · Shobhit Pandey · Shikhar Singh

**A1** END-TO-END SYSTEM · AUDIO NEVER LEAVES THE BANK'S NETWORK, AND THE MODEL NEVER EXECUTES AN ACTION




### INPUTS & PROVENANCE

Call audio (bank's own leg). Call context: requested action, caller no. vs registered, SIM-swap flag, repeat attempts. App credential registry.

### CONSENT & AUTHORITY

Bank's call-recording notice and processing basis. Evaluation uses consented or public clips only. Not collected: voiceprints, transcripts, phone data.

### OUTPUTS

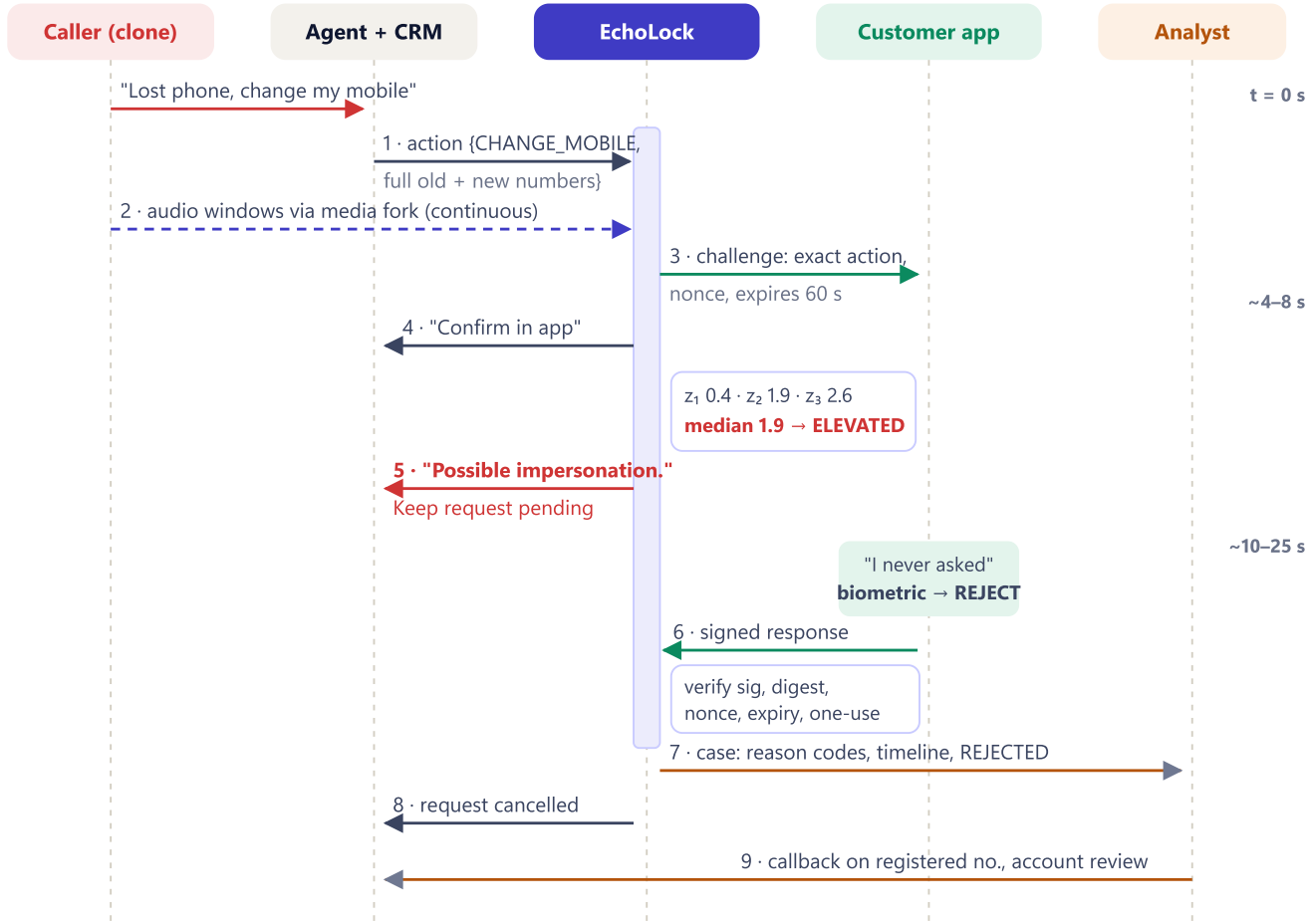
Voice state (not elevated / unknown / elevated), reason codes, challenge result, analyst case, and "eligible for bank checks".

### INTERFACES

Media WebSocket · internal gRPC/REST policy API · push + signed response · RBAC analyst UI · bank action API.

# Bind approval to the exact action, not to how a voice sounds.

## SEQUENCE · CLONED CALLER ASKS TO CHANGE THE REGISTERED MOBILE




## DETECTION PARAMETERS (TUNED, THEN FROZEN)

| | |
|---|---|
| **Window** | 4 s voiced speech, 2 s stride, 16 kHz input |
| **Observation** | $z_t$ = fake_logit − real_logit, with model revision |
| **Elevate when** | median(z of last 3 valid windows) ≥ $\tau_{codec}$, then latch. τ set at 1% dev false alarms (G.711 −2.27, AMR-NB 0.58), then frozen |
| **UNKNOWN** | <4 s speech, unsupported codec, clipping, or timeout |
| **Never** | treat softmax as fraud probability, or auto-act on it |
| **States** | REQUESTED → AWAITING_CONFIRM → COOLING_OFF (24 h for contact & limit changes) → ELIGIBLE · REVIEW · CANCELLED |

## SIGNED CONFIRMATION CONTRACT

```
{ "domain": "echolock.action-confirm.v1",
  "challenge_id": "srv-generated",
  "customer_id": "bank-ref",  "session_id": "call-ref",
  "credential_id": "pre-registered-key",
  "action_digest": "sha256(customer_ref | CHANGE_MOBILE
      | 9876503210 | 9876548876 | v1 | requested_at)",
  "nonce": "32 random bytes",
  "expires_at": 1791220000000,
  "decision": "reject" }
```

The digest covers the **full, unmasked values**. Only the display masks the old number; the app shows the full new number. Changed action → new challenge. Late, replayed or altered → rejected.

# Make escalation, evidence and failure handling explicit.

## ESCALATION PATH

- **Agent:** normal flow + app confirmation for every sensitive change
- **Unknown:** confirmation or registered callback, flagged "no voice evidence"
- **No app / phone lost / SIM swapped:** protect only. Re-register via branch, video-KYC or net banking + 24 h hold
- **Elevated:** request held, case queued by voice state × call context × impact
- **Analyst:** callback, then release / cancel with a written reason
- **Bank authoriser:** existing auth executes. Confirmed fraud goes into the bank's fraud reporting; customer guided to 1930 / cybercrime.gov.in

## EVIDENCE PACK (WHAT THE ANALYST SEES)

**Case #EL-0427 · CHANGE_MOBILE** ELEVATED



**Reasons:** 3/3 windows above τ · narrowband G.711 · quality OK
**App:** REJECTED by registered device at 00:21
**Model:** wav2vec-small-AD @9a13264 · config v0.3

## SECURITY, PRIVACY & ADVERSARIAL MITIGATIONS

| RISK | CONTROL |
|---|---|
| **Unseen cloner / codec** | Codec, noise and replay stress tests. Layer 2 doesn't depend on audio. Leave-one-engine-out tests. |
| **Adversarial audio** | Quality gate, strict decoder limits, sandboxed media worker. UNKNOWN instead of a guess. |
| **Approval replay / tamper** | Domain-separated signature, action digest, nonce, server-side expiry, one-time use. |
| **Coached genuine customer** | 24 h cooling-off, SMS + app notifications to old contacts, one-tap cancel. "Is someone telling you to do this?" → review. Rate-limited prompts. |
| **Forged scores** | mTLS-authenticated workers. Caller can't submit scores. Stale events rejected. |
| **Evidence misuse** | RBAC, encrypted store, access audit log, 7-day TTL, no raw audio in cases. |
| **Bias & accessibility** | False-alarm audit per language, gender and channel. TTS users take the same non-voice path. |
| **Model drift** | Versioned configs, shadow-mode testing of new models, rollback without disabling auth. |

## PERFORMANCE ASSUMPTIONS

**Measured:** 22 ms per 4 s window (RTX 3050 GPU), 299 ms on CPU. Rest are targets.




**≈1 MB RAM** per live call (16 s × 16 kHz float32). Bounded queue: overload → UNKNOWN. App prompt runs in parallel, adding ~0 s to the agent flow.

A APPENDIX · EVIDENCE AND DISCLOSURES

# What we built and measured, and what we used.

Code, team recordings, per-clip scores and the prototype: **github.com/tusharsinghbisht/raksham-iitd-ps1**

**1,727**
clips scored: 842 real, 768 cloned (Hindi), 117 team

**110**
speakers: 38 real, 84 cloned, 2 teammates

**6**
phone channels per clip, 10,362 scores in total

**22 ms**
per 4 s window, laptop GPU (RTX 3050); 299 ms on CPU

**26 / 26**
prototype tests pass: confirm, replay, tamper, expiry, review

**94.4 M**
weights loaded by our own PyTorch loader, strict check

## DETECTOR RESULTS · HINDI TEST SET, 960 CLIPS, 70 SPEAKERS, SPEAKER-DISJOINT

| CHANNEL | EER CALL-LEVEL [95% CI] | T (DEV) | DETECTED @T | FALSE ALARM @T | FREEVC / XTTS | TEAM FALSE ALARM |
|---|---|---|---|---|---|---|
| **Clean 16 kHz** | **1.6%** [0.8%–2.7%] | +1.13 | 95.8% | 0.7% | 100% / 91% | 1.7% |
| **8 kHz narrowband** | **3.6%** [2.1%–5.1%] | -1.25 | 95.8% | 2.1% | 100% / 90% | 8.5% |
| **G.711** | **3.4%** [2.1%–5.0%] | -2.27 | 96.6% | 3.9% | 100% / 93% | 16.2% |
| **Opus 12k** | **5.9%** [3.5%–8.5%] | -1.57 | 93.9% | 5.3% | 100% / 87% | 17.1% |
| **AMR-NB 12.2k** | **8.5%** [5.4%–11.8%] | +0.58 | 83.8% | 3.5% | 90% / 76% | 4.3% |
| **G.711 + 3% loss + noise** | **21.7%** [17.0%–25.7%] | +0.43 | 26.0% | 0.7% | 28% / 24% | 7.7% |

Kathbath real vs IndicSynth clones. τ fixed on a dev split at 1% false alarms, never tuned on test. Same-person check (14 speakers): 2.3% clean, 2.3% G.711.

## WHAT WE LEARNED · PROBLEMS WE FIXED

1. **Loss + noise breaks detection** (26% caught). Poor audio → UNKNOWN; app confirmation never relies on the score.
2. **τ shifts -2.27 → +1.13** by channel, so it's set per codec, and re-set on each bank's calls in shadow mode (our voices: 16% false alarms on G.711).
3. **fairseq won't install** on Windows, so we rewrote wav2vec 2.0 in PyTorch. Excluded FLEURS and MLAAD (in the model's training set).
4. **IndicSynth shards are sorted** (first pull: 5 XTTS speakers), so we sampled at random offsets → 84. Split by speaker ID across both datasets.

## DISCLOSURES

**AI-generated content:** Claude (Anthropic, via Claude Code) helped write and lay out this deck, the logo, the recording script and our evaluation code. An earlier plan was drafted with Claude; a first review and the workflow prototype with an OpenAI coding assistant. The team reviewed all of it and owns every claim. All diagrams are inline SVG/HTML; no generated images, stock photos or templates.

**Model:** NII wav2vec-small-anti-deepfake, Hugging Face rev 9a13264, on facebook/wav2vec2-base. CC BY-NC-SA 4.0, research only. Published weights, no fine-tuning. MMS-1B and AASIST-L are cited only, not run.

**Datasets:** Kathbath Hindi (AI4Bharat, CC BY 4.0), real speech. IndicSynth Hindi (ACL 2025, CC BY-NC 4.0), XTTS-v2 and FreeVC clones. Recordings of 2 team members, released with their consent at github.com/tusharsinghbisht/raksham-iitd-ps1. No bank, customer or platform data. We have cloned no one yet; demo clones will use consenting teammates only.

**Assets & software:** IIT Delhi emblem and Amazon logo (Wikimedia Commons), trademarks of their owners, used only to name the organisers. Segoe UI, PyTorch, SciPy, FFmpeg, Hugging Face Hub, Microsoft Edge.

**Sources:** McAfee, *The Artificial Imposter* (2023); Android CAPTURE_AUDIO_OUTPUT docs and Google Play policy (May 2022); RBI Authentication Directions 2025; DPDP Act 2023; TRAI 1600-series; NII model card and arXiv:2506.21090; RFC 7866 (SIPREC).

**Status:** slide 8, A2 τ values and this page are measured by us (12 Oct 2026). Slide 7 chart, app screens, evidence pack and cost are illustrative; prototype voice scores are simulated.